



ALLIANCE
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Bachelor of Computer Applications (Hons. Game Development)
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Subject: Database Management System

Topic: ER Diagram Design and Relational Schema

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ER Diagram and Relational Schema

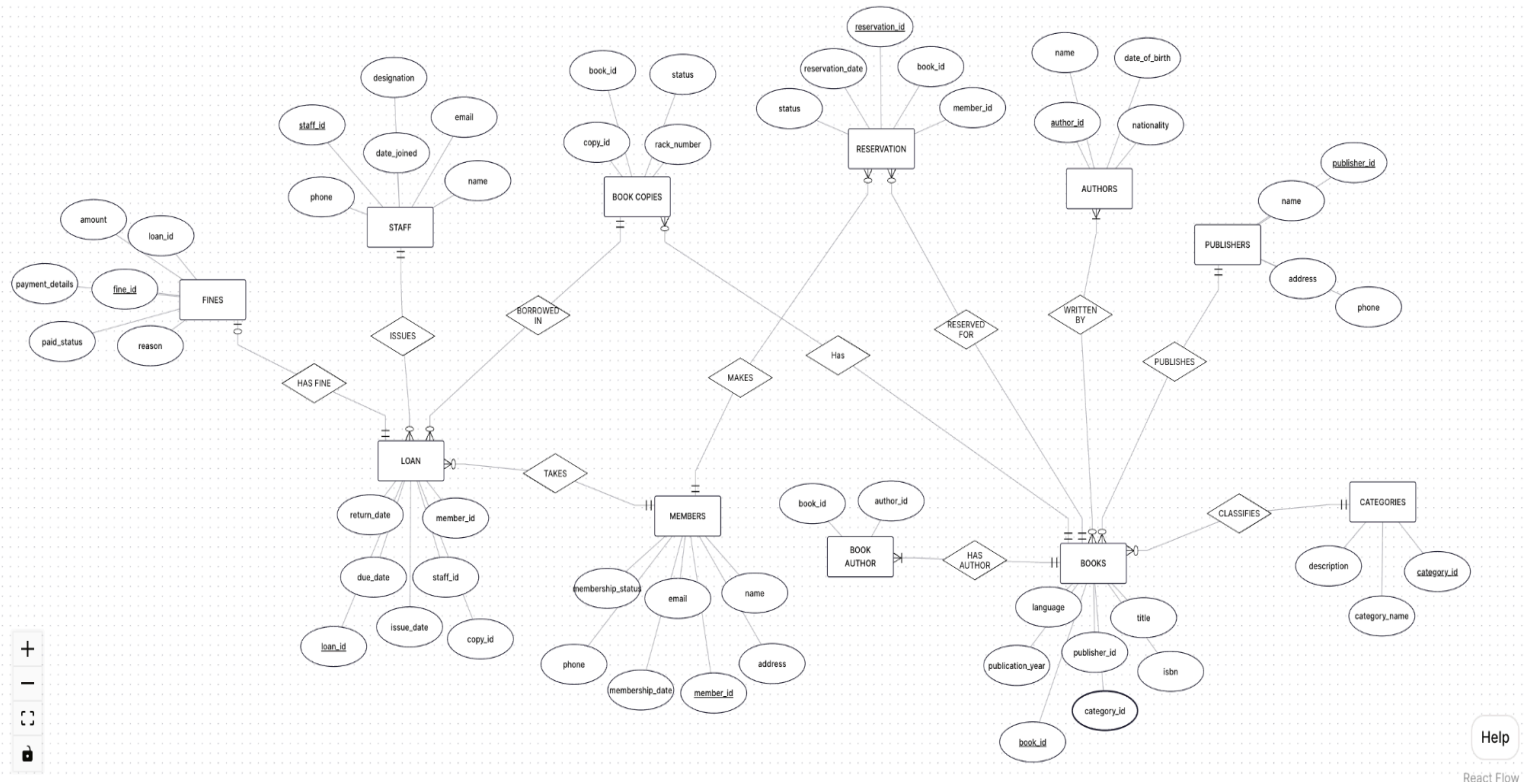
1. List of Entities and Attributes

Primary keys are shown in bold. Foreign keys are marked (FK).

No.	Entity	Attributes
1	MEMBERS	member_id (PK) , name, email, phone, address, membership_date, membership_status
2	STAFF	staff_id (PK) , name, email, phone, designation, date_joined
3	BOOKS	book_id (PK) , title, isbn, publication_year, language, publisher_id (FK), category_id (FK)
4	BOOK COPIES	copy_id (PK) , rack_number, status, book_id (FK)
5	AUTHORS	author_id (PK) , name, nationality, date_of_birth
6	PUBLISHERS	publisher_id (PK) , name, phone, address
7	CATEGORIES	category_id (PK) , category_name, description
8	LOAN	loan_id (PK) , issue_date, due_date, return_date, copy_id (FK), member_id (FK), staff_id (FK)
9	FINES	fine_id (PK) , amount, reason, paid_status, payment_details, loan_id (FK)
10	RESERVATION	reservation_id (PK) , reservation_date, status, book_id (FK), member_id (FK)
11	BOOK AUTHOR	book_id (PK, FK) , author_id (PK, FK)

2. ER Diagram

Diagram name: Library Management System



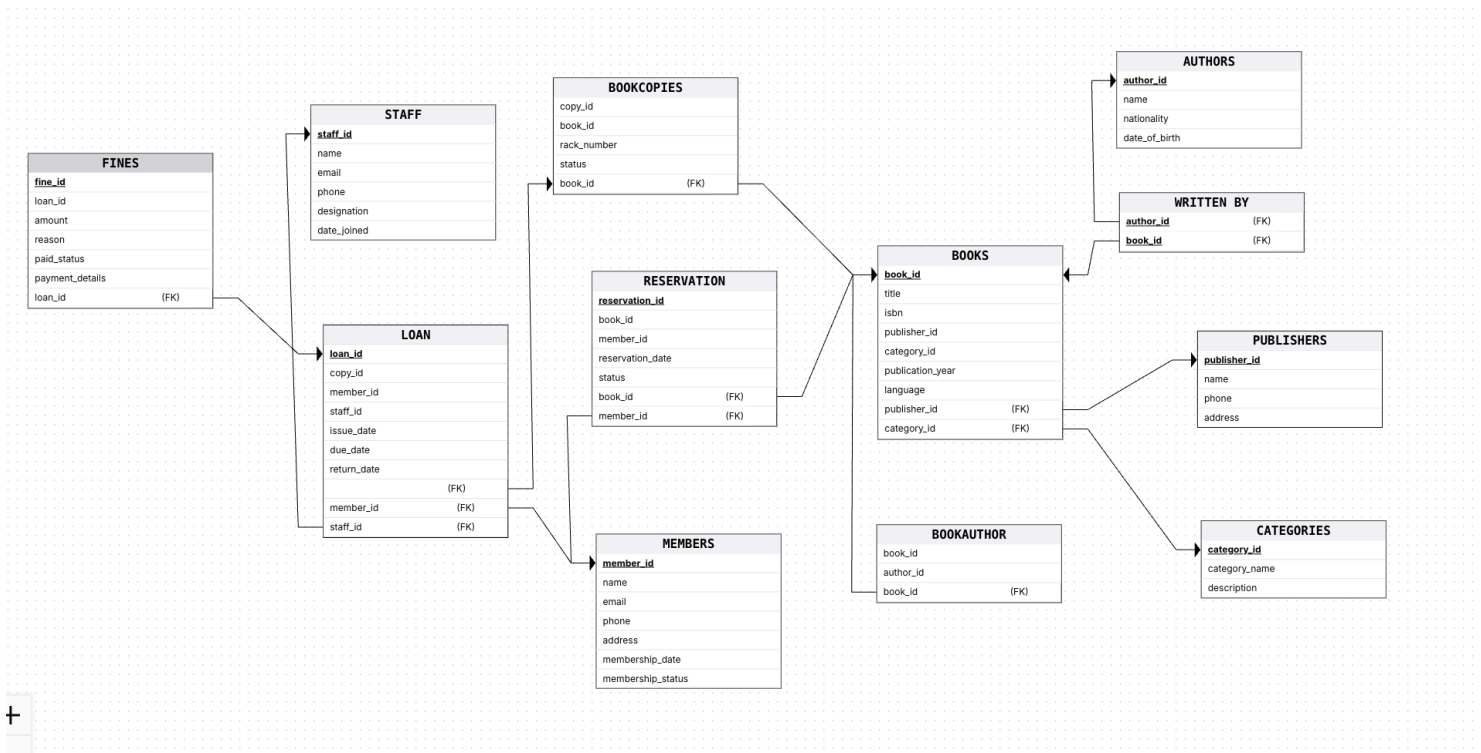
Notation:

- **rectangles** are entities,
- **ellipses** are attributes (underlined = primary key),
- **diamonds** are relationships,
- **crow's-foot** symbols show cardinality.

3. Relationships and Cardinalities

Relationship	Entity 1	Entity 2	Cardinality
ISSUES	STAFF	LOAN	One-to-Many (1:M)
BORROWED IN	BOOK COPIES	LOAN	One-to-Many (1:M)
TAKES	MEMBERS	LOAN	One-to-Many (1:M)
HAS FINE	LOAN	FINES	One-to-One (1:1); a fine is optional
MAKES	MEMBERS	RESERVATION	One-to-Many (1:M)
RESERVED FOR	BOOKS	RESERVATION	One-to-Many (1:M)
HAS	BOOKS	BOOK COPIES	One-to-Many (1:M)
PUBLISHES	PUBLISHERS	BOOKS	One-to-Many (1:M)
CLASSIFIES	CATEGORIES	BOOKS	One-to-Many (1:M)
WRITTEN BY / HAS AUTHOR	BOOKS	AUTHORS	Many-to-Many (M:N)

4. Relational Schema



5. Relations (Table)

Table	Attributes
MEMBERS	member_id , name, email, phone, address, membership_date, membership_status
STAFF	staff_id , name, email, phone, designation, date_joined
PUBLISHERS	publisher_id , name, phone, address
CATEGORIES	category_id , category_name, description
AUTHORS	author_id , name, nationality, date_of_birth
BOOKS	book_id , title, isbn, publication_year, language, publisher_id, category_id
BOOK_AUTHOR	book_id , author_id
BOOK_COPIES	copy_id , rack_number, status, book_id
LOAN	loan_id , issue_date, due_date, return_date, copy_id, member_id, staff_id
FINES	fine_id , amount, reason, paid_status, payment_details, loan_id
RESERVATION	reservation_id , reservation_date, status, book_id, member_id

Bold attributes form the primary key of each table.

Conversion rules used

- Each entity became a table, and its key attribute became the primary key.
- For every **1:M** relationship, the primary key of the “one” side was added as a foreign key in the “many” side table.
- The **1:1** relationship between LOAN and FINES places loan_id in FINES as a unique foreign key.
- The **M:N** relationship between BOOKS and AUTHORS became the associative table BOOK_AUTHOR, whose composite primary key is made of the two foreign keys.

6. Primary and Foreign Key Identification

Table	Primary Key	Foreign Key	References
MEMBERS	member_id	–	–
STAFF	staff_id	–	–
PUBLISHERS	publisher_id	–	–
CATEGORIES	category_id	–	–
AUTHORS	author_id	–	–
BOOKS	book_id	publisher_id	PUBLISHERS(publisher_id)
		category_id	CATEGORIES(category_id)
BOOK_AUTHOR	book_id, author_id (composite)	book_id	BOOKS(book_id)
		author_id	AUTHORS(author_id)
BOOK_COPIES	copy_id	book_id	BOOKS(book_id)
LOAN	loan_id	copy_id	BOOK_COPIES(copy_id)
		member_id	MEMBERS(member_id)
		staff_id	STAFF(staff_id)
FINES	fine_id	loan_id (unique)	LOAN(loan_id)
RESERVATION	reservation_id	book_id	BOOKS(book_id)
		member_id	MEMBERS(member_id)

7. Brief Relationship Explanation

- **STAFF – LOAN (ISSUES, 1:M):**
One staff member issues many loans, but exactly one staff member issues each loan.
- **BOOK COPIES – LOAN (BORROWED IN, 1:M):**
A physical copy can be lent many times over time, but each loan record is for one copy.
- **MEMBERS – LOAN (TAKES, 1:M):**
A member can take many loans, while each loan belongs to one member.

- **LOAN – FINES (HAS FINE, 1:1):**
A loan produces at most one fine (for example, for a late return), and each fine belongs to one loan.
- **MEMBERS – RESERVATION (MAKES, 1:M):**
A member can make many reservations; each reservation is made by one member.
- **BOOKS – RESERVATION (RESERVED FOR, 1:M):**
A book title can be reserved many times by different members, so each reservation is for one title.
- **BOOKS – BOOK COPIES (HAS, 1:M):**
A title can have many physical copies on the shelves, and each copy belongs to one title.
- **PUBLISHERS – BOOKS (PUBLISHES, 1:M):**
A publisher publishes many books; each book has one publisher.
- **CATEGORIES – BOOKS (CLASSIFIES, 1:M):**
A category groups many books, and each book belongs to one category.
- **BOOKS – AUTHORS (WRITTEN BY, M:N):**
A book can have several authors, and an author can write several books. This is resolved by the BOOK_AUTHOR table.